Paolo Nessim

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EDUCATION

Montgomery College — Germantown, MD

A.S. Computer Engineering and A.A. Computer Science — GPA 3.58 / 4.0 — *Expected May 2027*

- Dean's List; Phi Theta Kappa Honor Society; Southern Management Leadership Program Scholar.
- Coursework: Computer Science I–II (Java), Calculus I–II, General Chemistry for Engineers, Intro to Engineering Design. In progress: Digital Logic Design, Discrete Structures, Physics I.
- Planned transfer, Fall 2027: University of Maryland, B.S. Cyber-Physical Systems Engineering at the Universities at Shady Grove (application in progress; expected B.S. May 2029).

EXPERIENCE

University of Maryland, A. James Clark School of Engineering — Rockville, MD (Shady Grove)

Research Intern, Cyber-Physical Systems Engineering (CPSE) — *Aug 2025 – Present*

- Built the real-time computer vision pipeline (Python, OpenCV) for Magic Gambit v2, a robotic chess system: camera feed to board state to robotic-arm motion plan, tracking every piece as it moves.
- Designed the system controller from scratch: a custom ESP32-WROOM-32E PCB in KiCad with USB-C programming, CH340C auto-reset, ESD protection, and a TPS61022 boost power stage. Zero ERC errors; parts sourced against LCSC for JLCPCB fabrication.
- Implemented an ESP-NOW wireless link to a RoArm-M2-S robotic arm and a live web interface for board-state visualization and control.

University of Maryland, CPSE Summer Program — Rockville, MD

Hardware/Software Engineer — *Jun 2025 – Aug 2025*

- Built ICantSee, an assistive device pairing live computer vision with haptic feedback so visually impaired users can sense obstacles in real time. Selected from the cohort to continue as an academic-year research intern.

NASA L'SPACE Mission Concept Academy — Remote

Computer Engineering Participant — *Jan 2026 – May 2026*

- Carried a space mission concept through Preliminary Design Review on a multidisciplinary team, writing NASA-standard systems, science, and operations documentation.

Nessimworks — Clarksburg, MD

Founder, Full-Stack Developer — *Jan 2023 – Present*

- Scope, build, deploy, and maintain production websites and web apps for small-business and nonprofit clients (React, Next.js, PostgreSQL), owning requirements through delivery.

PROJECTS

Satgaze — Real-Time 3D Satellite Tracker — satgaze.com

- TypeScript, Next.js, Three.js/WebGL. Renders live positions of 24,000+ satellites in the browser, propagated from TLE data with SGP4 orbital mechanics.

OpenChess — Smart Chessboard Hardware github.com/PaoloJN/openchess

- KiCad, ESP32, C++ (PlatformIO), BLE, Expo. Chessboard that knows where every piece is: 64 hall-effect sensors, 81 RGB LEDs, OLED menu, printed case, and a BLE phone app, all on a custom ESP32 PCB. Plays online through Lichess. Automated validators check schematic wiring and placement before every fabrication run.

Sensei Puck — Submersible BLE Water Sensor github.com/PaoloJN/sensei-puck

- KiCad 10, nRF52840, C (PlatformIO), BLE. 46 mm sealed puck that logs conductivity, temperature, pressure, spectral, capacitive wetness, and audio from inside a water container. Complete, ERC-clean schematic; 4-layer, 139-footprint board; compiling bring-up firmware and OTA plan. Archived before fabrication with a written decision log on why.

kicad-ai-tools — Agent-Driven PCB Design github.com/PaoloJN/kicad-ai-tools

- Python, KiCad 10 IPC API. Tooling that lets an AI agent edit a live KiCad design over IPC with a propose, gate, accept loop, using kicad-cli as the single source of truth for DRC and ERC.

Beacon — Self-Hosted Error Tracking Platform github.com/PaoloJN/beacon

- Python, React, Docker. Lightweight Sentry alternative: Python ingestion core and REST API covered by 49 tests, Python and Node.js SDKs, React triage dashboard, alerting, one Docker image.

Recap — AI for YouTube, Open Source, 660+ GitHub stars github.com/PaoloJN/recap

- TypeScript, React, Plasmo, OpenAI / Anthropic / Google APIs. Chrome extension that adds summaries, a searchable transcript, and chat with citations that seek the video, in a sidebar. Bring-your-own-key, no server, no telemetry.

Snipprompt — Chrome Extension SaaS snipprompt.com

- Plasmo (MV3), Next.js, Stripe, Neon Postgres. Prompt-snippet manager live on the Chrome Web Store with paying subscribers. Built, shipped, and operated solo, including a native macOS menubar companion (Swift) and a Raycast extension.

SKILLS

Languages: C/C++ (embedded), Python, TypeScript/JavaScript, Java, SQL

Embedded & Hardware: KiCad schematic capture and PCB layout, ESP32 and nRF52840 (ESP-IDF, PlatformIO, Arduino), ESP-NOW, BLE, UART/I2C/SPI, sensor integration (hall-effect, haptic, camera), power design (boost regulators, USB-C), oscilloscopes and bench debugging, digital logic, Fusion 360, Onshape

Software & Tools: OpenCV, computer vision, ROS, real-time systems, REST APIs, Docker, Linux, Git, PostgreSQL, React, Next.js, Node.js, MCP servers and LLM agent tooling, automated testing, technical documentation

Clearance: U.S. citizen, eligible for a U.S. security clearance (none held)

LEADERSHIP & ACTIVITIES

Co-Lead, AI Project Team, Raptors Who Code, Montgomery College (Feb 2024 – Present). Member, MC AI Club. Open-source maintainer: 35 public repositories, ~1,250 GitHub contributions in 2026.